

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q21-1844

Moser Generating Station 138 kV
December 04, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1844. The Interconnection Customer proposes a 67.7 MW solar plus storage facility interconnecting at the Moser Generating Station 138 kV substation in the Pennsylvania area. The request was received on August 01, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 585 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$5,179,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Demar / Moser Generating Sta 138 kV	Owatonna / Moser Generating Station Substation 345 kV	117.9	272	44.0
Delmar / Delta Uppco 138 kV	Owatonna / Moser Generating Sta 345 kV line	111.6	267	44.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Delmar / Moser Generating Sta 138 kV	Rebuild line with higher-capacity conductor	\$788,000	92% (\$722,000)
Delmar / Delta Uppco 138 kV	Rebuild line with higher-capacity conductor	\$4,865,000	92% (\$4,457,000)
	Total allocated to project		\$5,179,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.